RReLU#

https://pytorch.org/docs/stable/generated/torch.nn.modules.activation.RReLU.

Description:

Applies the randomized leaky rectified linear unit function, element-wise. Method described in the paper: Empirical Evaluation of Rectified Activations in Convolutional Network. The function is defined as: where aaa is randomly sampled from uniform distribution U(lower,upper)\mathcal{U}(\text{lower}, \text{upper})U(lower,upper) during training while during evaluation aaa is fixed with a=lower+upper2a = \frac{\text{lower} + \text{upper}}{2}a=2lower+upper. lower (float) – lower bound of the uniform distribution. Default: 18\frac{1}{8}81

Function Signature

```
class torch.nn.modules.activation.RReLU(lower=0.125,
upper=0.3333333333333333, inplace=False)[source]#

Parameters

Shape:

extra_repr()[source]#

Return type
```

Returns:

Tensor

Code Examples

```
>>> m = nn.RReLU(0.1, 0.3)
>>> input = torch.randn(2)
>>> output = m(input)
```

Detailed Documentation

Documentation

Applies the randomized leaky rectified linear unit function, element-wise.

Method described in the paper: Empirical Evaluation of Rectified Activations in Convolutional Network.

The function is defined as:

where aaa is randomly sampled from uniform distribution U(lower,upper)\mathcal{U}(\text{lower}, \text{upper})U(lower,upper) during training while during evaluation aaa is fixed with a=lower+upper2a = \frac{\text{lower} + \text{upper}}{2}a=2lower+upper.

lower (float) – lower bound of the uniform distribution. Default: 18\frac{1}{8}81

upper (float) – upper bound of the uniform distribution. Default: 13\frac{1}{3}31

inplace (bool) – can optionally do the operation in-place. Default: False

Input: (∗)(*)(∗), where ∗*∗ means any number of dimensions.

Output: (∗)(*)(∗), same shape as the input.

Return the extra representation of the module.

Runs the forward pass.